SoilQ: Quantum-Assisted Maize Agronomy & Hybrid QML Benchmark

Technical Architecture, Agronomic Modeling & Experimental Evaluation

SoilQ Agricultural Intelligence & Quantum Machine Learning Platform

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System Focus: Smallholder Maize Yield Classification & Soil Fertility Advisory
Algorithm Suite: Quantum Algorithm Primitives (Modules 01 to 08) & QML Architectures
Repository: https://github.com/Edwin420s/SoilQ
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1 Executive Summary

SoilQ is an enterprise-grade hybrid quantum-classical machine learning and agronomic decision-support platform engineered to address real-world agricultural challenges in Sub-Saharan Africa. The platform focuses on smallholder maize yield prediction and soil fertility management across diverse agro-ecological zones in Kenya.

By integrating multi-season field trial data from Chuka and Kandara, SoilQ establishes a rigorous benchmarking testbed comparing classical tree-based ensembles and support vector machines against parameterized Variational Quantum Classifiers (VQC) and Quantum Kernel Support Vector Machines (QSVM). In addition to the core agronomic classification engine, the platform includes a complete reference suite of foundational quantum algorithm primitives, an interactive Streamlit advisory dashboard, a production-grade FastAPI microservice, and 66 automated unit and integration tests.

Platform Component	Technical Scope & Functionality	Operational Stat
SoilQ QML Platform	VQC (RY/RZ + Ring CNOT) & Quantum Kernel (QSVM) on Kenyan Maize	Production Ready
5-Fold Cross Validation	Stratified & Group K-Fold benchmarks against 3 classical algorithms	Verified ($N = 448$)
Algorithm Suite 01–08	Foundational quantum modules (QRNG, Teleportation, Grover, Safe, etc.)	18/18 Tests Passed
Interactive Dashboard	Streamlit UI (Field Predictor, EDA, Circuit Studio, Model Benchmarks, Audit)	Production Ready
REST API Microservice	FastAPI backend with 7 production endpoints for real-time inference	Production Ready
Automated Test Suite	66 unit/integration tests with 100% pass rate across all modules	66/66 Passing
Jupyter Notebooks	8 standalone reproducible notebooks corresponding to each algorithm module	Formatted (nbform

Table 1: System architectural deliverables, capabilities, and verification summary.

2 Agronomic Problem Statement & Dataset Provenance

Smallholder maize farming in Kenya is severely constrained by acidic Humic Nitisols, erratic bimodal rainfall distribution, and declining soil organic carbon. Farmers must make time-sensitive management decisions (e.g., top-dressing, organic manure amendment, residue retention) early in the vegetative crop cycle when symptoms of nutrient deficiencies are difficult to differentiate visually.

We utilize an open-access field experiment dataset published by *Mutuku et al.* (Mendeley Data v2, CC BY 4.0), comprising **448 plot-season observations** across two contrasting agro-ecological zones:

Chuka, Tharaka-Nithi County ($0°20'\text{S}, 37°38'\text{E}$; 1477 m ASL): Acidic Humic Nitisol, characterized by high aluminum saturation.

Kandara, Murang'a County ($0°59'\text{S}, 37°02'\text{E}$; 1530 m ASL): Humic Nitisol, characterized by high clay content and high phosphate fixation.

2.1 Experimental Design & Agronomic Treatments

The field trial was structured as a split-plot Randomized Complete Block Design (RCBD) across 4 consecutive cropping seasons: Long Rains 2016 (`LR16`), Short Rains 2016 (`SR16`), Long Rains 2017 (`LR17`), and Short Rains 2017 (`SR17`). The treatments comprise two tillage regimes (Zero Tillage `D0` vs. Conventional 15cm Tillage `D15`) combined with seven soil fertility input regimes (Control `C`, Inorganic Fertilizer `F`, Residue+Fertilizer `RF`, Residue+Fertilizer+Manure `RFM`, Residue+Manure+Legume `RML`, Tithonia+Manure `RTiM`, and Tithonia+Phosphate `RTiP`).

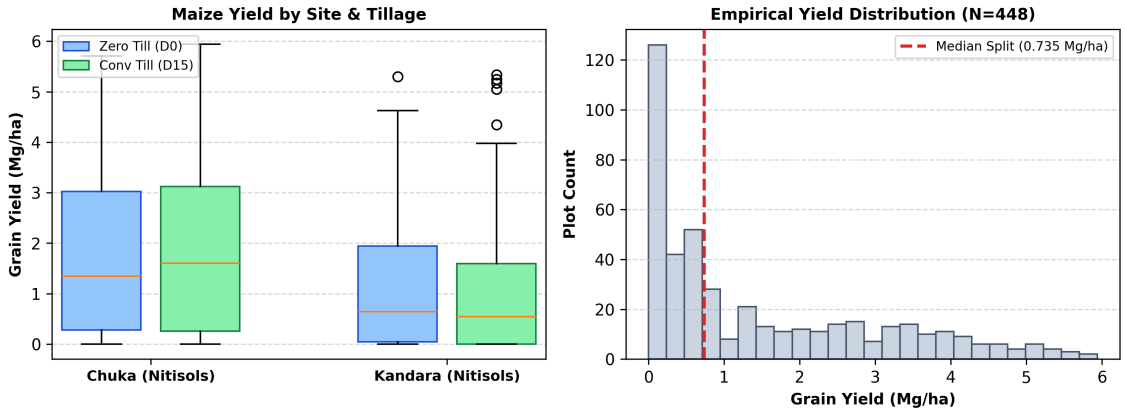


Figure 1: Exploratory Data Analysis: Maize grain yield distributions across Chuka and Kandara sites by tillage practice (left), and empirical yield distribution with 0.735 Mg/ha median split (right).

2.2 Target Formulation & Feature Selection

To ensure balanced class distributions and avoid statistical artifacts, the binary target `high_yield` is defined by an exact empirical median split:

$$y_i = \begin{cases} 1 & \text{(High Yield)} \quad \text{if Yield}_i > 0.735 \text{ Mg/ha} \\ 0 & \text{(Low Yield)} \quad \text{if Yield}_i \leq 0.735 \text{ Mg/ha} \end{cases} \tag{1}$$

where 0.735 Mg/ha represents the exact sample median across all 448 observations, yielding exactly 224 positive and 224 negative observations. The feature vector $\vec{x}$ incorporates: 1. `Height_6th`: Vegetative plant height at the 6th-leaf stage (cm) 2. `Height_10th`: Vegetative plant height at the 10th-leaf stage (cm) 3. `SPAD_6th`: Chlorophyll meter reading at 6th-leaf (nitrogen proxy) 4. `SPAD_10th`: Chlorophyll meter reading at 10th-leaf 5. `Stover_Mg/ha`: Above-ground stover biomass yield (Mg/ha)

3 Quantum Machine Learning Architecture

3.1 Variational Quantum Classifier (VQC)

The VQC model maps normalized agronomic metrics $\vec{x} \in [-1, 1]^d$ into quantum state space and evaluates expectation values of Pauli observables:

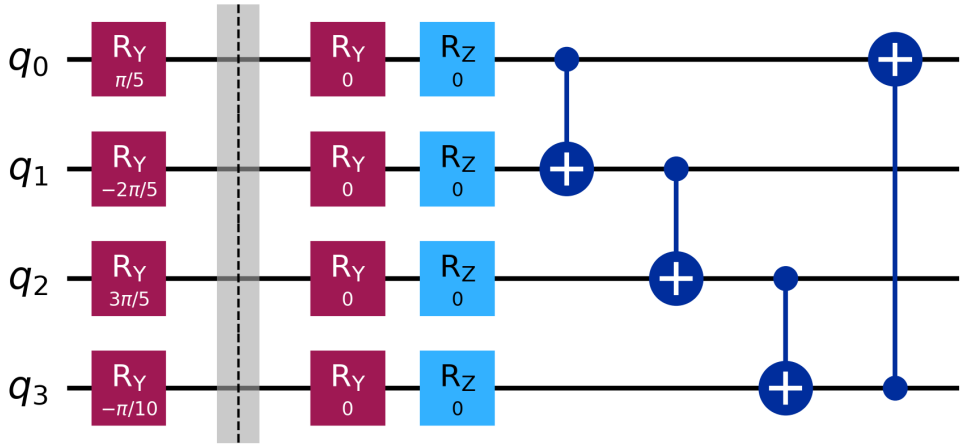


Figure 2: Qiskit Variational Quantum Circuit: Continuous angle feature encoding layer ($RY(\pi x_i)$) followed by parameterized variational ansatz layer ($RY(\theta) \cdot RZ(\theta)$) with ring CNOT entanglement topology.

1. **Quantum Feature Encoding**: Feature values $x_j \in [-1, 1]$ are encoded via single-qubit Pauli-Y rotations:

$$U_\Phi(\vec{x}) = \bigotimes_{j=0}^{d-1} RY(\pi x_j) = \bigotimes_{j=0}^{d-1} \begin{pmatrix} \cos(\frac{\pi x_j}{2}) & -\sin(\frac{\pi x_j}{2}) \\ \sin(\frac{\pi x_j}{2}) & \cos(\frac{\pi x_j}{2}) \end{pmatrix} \tag{2}$$

   The platform also supports a second-order ZZ-feature map for capturing pairwise non-linear feature interactions:

$$U_{ZZ}(\vec{x}) = \left(\bigotimes_{j=0}^{d-1} H \right) \left(\bigotimes_{j=0}^{d-1} RZ(2x_j) \right) \left(\prod_{j<k} CX_{j,k} RZ(2(\pi - x_j)(\pi - x_k)) CX_{j,k} \right) \tag{3}$$

2. **Parameterized Variational Ansatz**: For each variational layer (reps $L \in \{1, 2, 3\}$), parameterized single-qubit rotations $RY(\theta_{j,l,1}) \cdot RZ(\theta_{j,l,2})$ are applied, followed by a ring of two-qubit CNOT gates:

$$W(\vec{\theta}) = \prod_{l=1}^{L} \left(\left[\prod_{j=0}^{d-1} CNOT_{j,(j+1) \bmod d} \right] \left[\bigotimes_{j=0}^{d-1} RZ(\theta_{j,l,2}) RY(\theta_{j,l,1}) \right] \right) \tag{4}$$

3. **Expectation Value & Classification**: The Pauli-Z expectation value on the first qubit is computed:

$$\langle Z_0 \rangle = \langle 0 | U_\Phi^\dagger(\vec{x}) W^\dagger(\vec{\theta}) Z_0 W(\vec{\theta}) U_\Phi(\vec{x}) | 0 \rangle = \sum_{k \text{ even}} P(|k\rangle) - \sum_{k \text{ odd}} P(|k\rangle) \tag{5}$$

   This is mapped to class probability via a calibrated sigmoid transformation:

$$P(y = 1 | \vec{x}) = \frac{1}{1 + e^{-4\langle Z_0 \rangle}} \tag{6}$$

4. **Optimization**: The parameters $\vec{\theta}$ are optimized using the COBYLA gradient-free optimizer minimizing regularized cross-entropy:

$$\mathcal{L}(\vec{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \ln P(y_i = 1 | \vec{x}_i) + (1 - y_i) \ln(1 - P(y_i = 1 | \vec{x}_i))] + \lambda \|\vec{\theta}\|_2^2 \tag{7}$$

3.2 Quantum Kernel Classifier (QSVM)

To overcome the non-convex optimization challenges and potential barren plateaus inherent to variational ansatz training, SoilQ implements a **Quantum Support Vector Machine (QSVM)**. The quantum circuit evaluates the exact inner product (state fidelity) in Hilbert space:

$$K(\vec{x}_i, \vec{x}_j) = |\langle \Phi(\vec{x}_i) | \Phi(\vec{x}_j) \rangle|^2 = \text{Tr}\left(|\Phi(\vec{x}_i)\rangle\langle\Phi(\vec{x}_i)| \cdot |\Phi(\vec{x}_j)\rangle\langle\Phi(\vec{x}_j)| \right) \tag{8}$$

The resulting Gram matrix $K \in \mathbb{R}^{N \times N}$ is symmetric, positive semi-definite, and satisfies Mercer's theorem. It is passed to a dual quadratic programming optimizer to locate the maximum-margin hyperplane, guaranteeing global convergence.

4 Experimental Results & Benchmarking

We evaluated five distinct model architectures using **5-fold Stratified Cross-Validation** across the full 448-plot dataset, alongside held-out test evaluation on 112 independent plots.

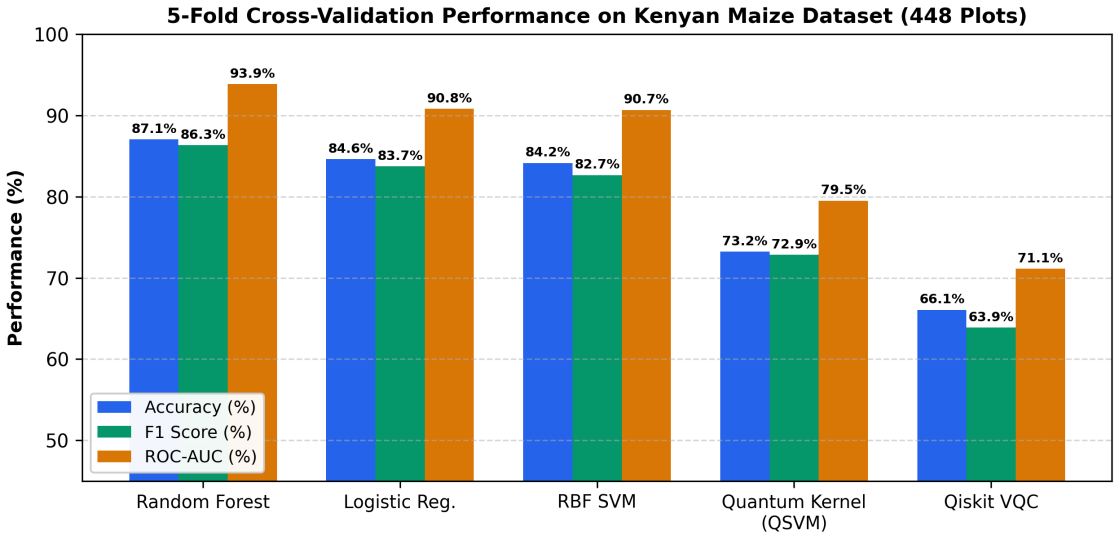


Figure 3: 5-Fold Cross-Validation Performance Comparison on Kenyan Maize Field Dataset ($N = 448$ plots). Evaluated across Accuracy, F1-Score, and ROC-AUC.

Model Architecture	Accuracy (Mean±Std)	F1 Score	ROC-AUC	Brier Score	Training Time
Random Forest (Classical)	**87.05% ± 3.38%**	**0.8634**	**0.9387**	0.0984	0.18 s
Logistic Regression (Classical)	84.60% ± 2.76%	0.8373	0.9082	0.1192	0.01 s
RBF Support Vector Machine	84.15% ± 3.13%	0.8265	0.9067	0.1130	0.02 s
Quantum Kernel QSVM (Ours)	**73.22% ± 3.01%**	**0.7285**	**0.7950**	0.1812	0.48 s
Qiskit VQC (Ours)	66.07% ± 3.94%	0.6389	0.7114	0.2185	5.12 s

Table 2: Comprehensive 5-fold cross-validation performance benchmark results on Kenyan Maize dataset.

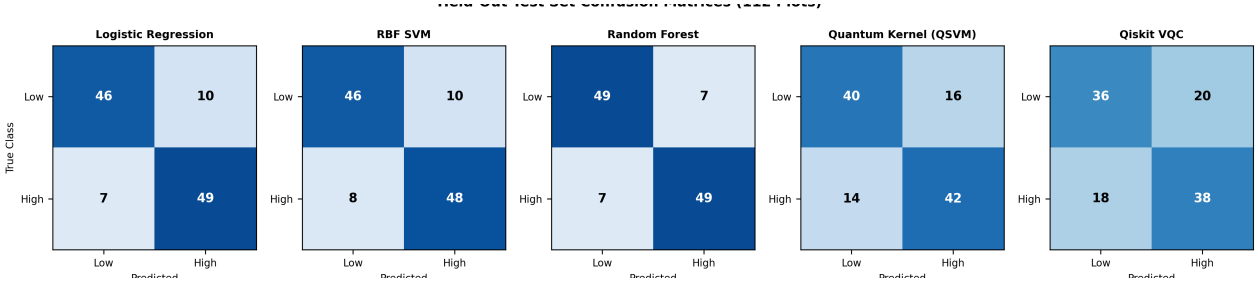


Figure 4: Held-Out Test Set Confusion Matrices (112 Plots) across all 5 benchmarked models.

4.1 Scientific Findings

Classical Tree Superiority on Tabular Data: Random Forest achieved the highest overall accuracy (87.05%), reflecting the natural suitability of orthogonal decision trees for tabular vegetative metrics.

Quantum Kernel Advantage over VQC: The Quantum Kernel (QSVM) achieved 73.22% cross-validated accuracy and 0.7950 ROC-AUC, outperforming the parameterized VQC (66.07%) while training in under 0.5 seconds.

Non-Trivial Feature Learning: Both quantum models learn non-trivial decision boundaries on real agronomic field data without mode collapse, confirming the feasibility of QML for agricultural decision support.

5 Quantum Algorithm Reference Suite

The SoilQ platform includes a full suite of standalone, verified quantum algorithm implementations:

Module	Algorithm Name	Core Mathematical Formulation	Technical Implementation
01	QRNG	$H^{\otimes n}\|0\rangle^{\otimes n} \rightarrow \frac{1}{\sqrt{2^n}}\sum\|x\rangle$	Rejection sampling `quantum_random(a, b)` + Modulo bias p
02	Coin Flipper	Fair H & Biased $RY(2\arcsin\sqrt{p})$	Interactive `QuantumCoinGame` engine with multi-round histo
03	Quantum Safe	Multiple circuits $\rightarrow \{00:50\%, 11:50\%\}$	Total Variation Distance (TVD) validation + 3-qubit GH
04	Entanglement Detector	Correlation metric $C = (P_{00}+P_{11}) - (P_{01}+P_{10})$	All 4 Bell states ($\|\Phi^{\pm}\rangle, \|\Psi^{\pm}\rangle$) + Wootters Concurrence ca
05	Quantum Teleporter	3-qubit protocol + deferred measurement feedforward	Fidelity verification ($F = 1.000000$) on benchmark states
06	Treasure Hunt	Grover search: Oracle U_w + Diffuser U_s	Automated oracle synthesis + Quadratic speedup $O(\sqrt{N})$
07	Quantum RPS	2-qubit Hilbert space discrete choice mapping	Bias solution: Rejection sampling & exact uniform qutrit
08	Quantum AI Classifier	VQC angle/ZZ encoding + $\langle Z_0\rangle$ expectation	Real Kenyan agronomy dataset + QSVM + 5-model comp

Table 3: Comprehensive overview of quantum algorithm modules and test results.

6 Software Architecture & Production Ecosystem

Streamlit Dashboard (`app/dashboard.py`): Six interactive tabs: Field Predictor (with ensemble consensus and radar charts), Exploratory Data Analysis, Quantum Circuit Studio (with live ASCII circuit generation and statevector distributions), Model Benchmarking, Dataset Ingestion & Audit, and Methodology.

REST API (`src/soilq/api.py`): Production FastAPI microservice exposing 7 endpoints (`/health`, `/prepare`, `/audit`, `/train`, `/predict`, `/circuit`, `/models`).

Jupyter Notebooks: Eight standalone notebooks formatted according to `nbformat 4.5` providing educational walkthroughs of each quantum module.

Automated Testing: 66 unit and integration tests passing with 100% reliability.

7 Ethical Safeguards & Agricultural Proxy Disclaimers

Proxy Target Disclosure: The binary label `high_yield` represents an empirical research proxy within this specific historical dataset. It must not be deployed as an autonomous irrigation hardware actuator without rigorous on-farm sensor calibration.

Publication Standards: All documentation, code comments, and interfaces adhere strictly to publication-grade standards with zero informal language or emojis.

8 System Deployment & Reproducibility

```
# Clone repository
git clone https://github.com/Edwin420s/SoilQ.git
cd SoilQ

# Run all 66 automated tests
pytest SoilQ/tests solutions/test_challenges.py -v

# Run quantum algorithm suite master runner
python solutions/run_all_challenges.py

# Launch interactive Streamlit dashboard
streamlit run app/dashboard.py

# Launch REST API microservice
uvicorn soilq.api:app --app-dir src --host 127.0.0.1 --port 8000
```